

(1) LU für  $A = \begin{pmatrix} 0 & 2 & 1 & -1 \\ -2 & 3 & 0 & 2 \\ 0 & 0 & 0 & 1 \\ 2 & 1 & 1 & 0 \end{pmatrix}$

Lös  $Ax = b$ ,  $b = (4, 19, 7, 12)^T$

$$\begin{pmatrix} 0 & 2 & 1 & -1 \\ -2 & 3 & 0 & 2 \\ 0 & 0 & 0 & 1 \\ 2 & 1 & 1 & 0 \end{pmatrix}$$

$p_1 = 2$   
 $\Rightarrow \begin{pmatrix} -2 & 3 & 0 & 2 \\ 0 & 2 & 1 & -1 \\ 0 & 0 & 0 & 1 \\ 2 & 1 & 1 & 0 \end{pmatrix}$

$\Rightarrow \begin{pmatrix} -2 & 3 & 0 & 2 \\ 0 & 2 & 1 & -1 \\ 0 & 0 & 0 & 1 \\ -1 & 4 & 1 & 2 \end{pmatrix}$

$p_2 = 2$   
 $\Rightarrow$

$$\begin{pmatrix} -2 & 3 & 0 & 2 \\ 0 & 2 & 1 & -1 \\ 0 & 0 & 0 & 1 \\ -1 & 2 & -1 & 4 \end{pmatrix}$$

$p_3 = 4$   
 $\Rightarrow \begin{pmatrix} -2 & 3 & 0 & 2 \\ 0 & 2 & 1 & -1 \\ -1 & 2 & -1 & 4 \\ 0 & 0 & 0 & 1 \end{pmatrix}$

$\Rightarrow \begin{pmatrix} -2 & 3 & 0 & 2 \\ 0 & 2 & 1 & -1 \\ -1 & 2 & -1 & 4 \\ 0 & 0 & 0 & 1 \end{pmatrix}$

$\Rightarrow L = \begin{pmatrix} 1 & & & \\ 0 & 1 & & \\ -1 & 2 & 1 & \\ 0 & 0 & 0 & 1 \end{pmatrix}, U = \begin{pmatrix} -2 & 3 & 0 & 2 \\ 2 & 1 & -1 & \\ -1 & 4 & & \\ 1 & & & \end{pmatrix}, P = \begin{pmatrix} 2 \\ 2 \\ 4 \end{pmatrix}$

$$(2) \quad f(x, y) = \frac{x}{y}, \quad x = -50 \pm 10, \quad y = 100 \pm 10$$

a) max. abs. Fehler für  $f$

b) Abschätzung mit Fehlerformel

$$a) \quad f(x, y) = \frac{-50}{100} = -\frac{1}{2}$$

$$x \in [-60, -40], \quad y \in [90, 110]$$

$$\Rightarrow \partial_x f(x, y) = \frac{1}{y} > 0$$

$$\partial_y f(x, y) = -\frac{x}{y^2} > 0$$

$\Rightarrow f$  ist monoton auf  $[-60, -40] \times [90, 110]$

$\Rightarrow$  reicht Randwerte abzulesen:

$$f(-60, 90) = -\frac{60}{90} = -\frac{2}{3}$$

$$f(-40, 110) = -\frac{40}{110} = -\frac{4}{11}$$

$$e_a = \max \left( |f(-60, 90) - f(-50, 100)|, |f(-40, 110) - f(-50, 100)| \right)$$

$$= \max \left( \left| -\frac{2}{3} + \frac{1}{2} \right|, \left| -\frac{4}{11} + \frac{1}{2} \right| \right)$$

$$= \max \left( \frac{1}{6}, \frac{3}{22} \right)$$

$$= \frac{1}{6}$$

$$b) \quad \partial_x f(x, y) = \frac{1}{y}, \quad \partial_y f(x, y) = -\frac{x}{y^2}$$

$$\Rightarrow e_a \approx \left| \frac{1}{100} \right| \cdot 10 + \left| -\frac{-50}{10000} \right| \cdot 10 = \frac{1}{10} + \frac{1}{20} = \frac{3}{20}$$

(3)

$$\begin{pmatrix} 8 & 12 \\ -1 & 4 \end{pmatrix} x = \begin{pmatrix} 4 \\ 5 \end{pmatrix}$$

a) Jacobi:  $x_1, \dots, x_4$ ,  $x_0 = 0$

b)  $B_J$ ,  $\|B_J\|_\infty$ ,  $\rho(B_J)$ , Konvergenz

a) Jacobi

$$x_1^{(k+1)} = \frac{1}{8} (4 - 12 x_2^{(k)})$$

$$x_2^{(k+1)} = \frac{1}{4} (5 + x_1^{(k)})$$

| $x^{(0)}$ | $x^{(1)}$     | $x^{(2)}$       | $x^{(3)}$        | $x^{(4)}$        |
|-----------|---------------|-----------------|------------------|------------------|
| 0         | $\frac{1}{2}$ | $-\frac{11}{8}$ | $-\frac{25}{16}$ | $-\frac{55}{64}$ |
| 0         | $\frac{5}{4}$ | $\frac{11}{8}$  | $\frac{25}{32}$  | $\frac{55}{64}$  |

b)  $B_J = D^{-1} (E+F) = \begin{pmatrix} \frac{1}{8} & 0 \\ 0 & \frac{1}{4} \end{pmatrix} \begin{pmatrix} 0 & -12 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & -\frac{3}{2} \\ \frac{1}{4} & 0 \end{pmatrix}$

$$\|B_J\|_\infty = \max \left( \left| -\frac{3}{2} \right|, \left| \frac{1}{4} \right| \right) = \frac{3}{2}$$

Eigenwerte:  $0 = \det(B_J - \lambda I) = \det \begin{pmatrix} -\lambda & -\frac{3}{2} \\ \frac{1}{4} & -\lambda \end{pmatrix} = \lambda^2 + \frac{3}{8}$

$$\Rightarrow \lambda_{1,2} = \pm i \sqrt{\frac{3}{8}}$$

$$\Rightarrow \rho = \max(|\lambda_1|, |\lambda_2|) = \sqrt{\frac{3}{8}} \approx 0.61$$

$$\rho(B_J) < 1 \Rightarrow \text{konvergiert}$$

④  $f(x) = \ln(x) - \cos(x)$

a) hat NS auf  $[1, 2]$

b) Bisektion bis  $1$ -Länge  $\frac{1}{4}$

c) Newton ab Mitte bis 6 Stellen sehen

a)  $f(1) \approx -0.54$        $f(2) \approx 1.109$

$\Rightarrow$  verschiedenen Vorzeichen + stetig

$\Rightarrow \exists$  mindestens eine Nullstelle

b)  $[a_0, b_0] = [1, 2]$  ,  $x_0 = \frac{3}{2}$  ,  $f(\frac{3}{2}) \approx 0.3347$

$[a_1, b_1] = [1, \frac{3}{2}]$  ,  $x_1 = \frac{5}{4}$  ,  $f(\frac{5}{4}) \approx -0.09$

$[a_2, b_2] = [\frac{5}{4}, \frac{3}{2}]$  , Länge =  $\frac{1}{4}$

c)  $x_0 = \frac{\frac{5}{4} + \frac{3}{2}}{2} = \frac{11}{8} = 1.375$

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}$$

$$f'(x) = \frac{1}{x} + \sin(x)$$

$$\Rightarrow x_{k+1} = x_k - \frac{\ln(x_k) - \cos(x_k)}{\frac{1}{x_k} + \sin(x_k)}$$

$$x_1 \approx 1.30246253$$

$$x_2 \approx 1.30296398$$

$$x_3 \approx 1.30296400$$

        

6 führende Stellen

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|       |    |   |   |    |
|-------|----|---|---|----|
| $x_i$ | -2 | 0 | 1 | 2  |
| $y_i$ | 4  | 0 | 1 | -4 |

a) IP

b) Approx. mit  $p(x) = ax + bx^3$

a)

|       |       |                           |                                                    |
|-------|-------|---------------------------|----------------------------------------------------|
| $x_i$ | $y_i$ |                           |                                                    |
| -2    | 4     |                           |                                                    |
| 0     | 0     | $\frac{0-4}{0-(-2)} = -2$ |                                                    |
| 1     | 1     | $\frac{1-0}{1-0} = 1$     | $\frac{1-(-2)}{1-(-2)} = 1$                        |
| 2     | -4    | $\frac{-4-1}{2-1} = -5$   | $\frac{-5-1}{2-0} = -3$ $\frac{-3-1}{2-(-2)} = -1$ |

$$\begin{aligned} \Rightarrow p(x) &= 4 - 2(x+2) + (x+2)x - (x+2)x(x-1) \\ &= 4 - 2x - 4 + x^2 + 2x - x^3 - x^2 + 2x \\ &= 2x - x^3 \end{aligned}$$

b)

$$A = \begin{pmatrix} -2 & -8 \\ 0 & 0 \\ 1 & 1 \\ 2 & 8 \end{pmatrix} \quad b = \begin{pmatrix} 4 \\ 0 \\ 1 \\ -4 \end{pmatrix}$$

$$A^T A = \begin{pmatrix} -2 & 0 & 1 & 2 \\ -8 & 0 & 1 & 8 \end{pmatrix} \begin{pmatrix} -2 & -8 \\ 0 & 0 \\ 1 & 1 \\ 2 & 8 \end{pmatrix} = \begin{pmatrix} 9 & 33 \\ 33 & 129 \end{pmatrix}$$

$$A^T b = \begin{pmatrix} 9 & 33 \\ 33 & 129 \end{pmatrix} \begin{pmatrix} 4 \\ 0 \\ 1 \\ -4 \end{pmatrix} = \begin{pmatrix} -15 \\ -63 \end{pmatrix}$$

$$A^T A = A^T b \quad (\Leftrightarrow) \quad \begin{array}{cc|c} 9 & 33 & -15 \\ 33 & 129 & -63 \end{array}$$

$$(\Leftrightarrow) \quad \begin{array}{cc|c} 3 & 11 & -5 \\ 33 & 129 & -63 \end{array}$$

$$(\Leftrightarrow) \quad \begin{array}{cc|c} 3 & 11 & -5 \\ 0 & 8 & -8 \end{array}$$

$$(\Leftrightarrow) \quad b = -1, \quad a = 2$$

$$\Rightarrow p(x) = 2x - x^3$$



Alternative:

- Interpolationspolynom von a) ~~hat~~ ist

$$p(x) = 2x - x^3$$

$\Rightarrow$  Struktur punkt

$\Rightarrow 2x - x^3$  ist auch Bestappr.

⑥  $\int_{-2}^2 \sqrt{4-x^2} dx$

a) Simpson

b)  $G_2$

c) Richardson mit Ordnung

$$\begin{aligned} a) \quad \int(f) &= \frac{b-a}{6} (f(a) + 4f(\frac{a+b}{2}) + f(b)) \\ &= \frac{4}{6} (f(-2) + 4f(0) + f(2)) \\ &= \frac{4}{6} (0 + 4 \cdot 2 + 0) \\ &= \frac{16}{3} \end{aligned}$$

$$\begin{aligned} b) \quad \text{auf } [-1, 1] : \quad x: & \quad -\sqrt{\frac{3}{5}} \quad 0 \quad \sqrt{\frac{3}{5}} \\ \beta: & \quad \frac{5}{9} \quad \frac{8}{9} \quad \frac{5}{9} \end{aligned}$$

$$\begin{aligned} \text{auf } [-2, 2] : \quad x: & \quad -2\sqrt{\frac{3}{5}} \quad 0 \quad 2\sqrt{\frac{3}{5}} \\ & \quad \frac{10}{9} \quad \frac{16}{9} \quad \frac{10}{9} \end{aligned}$$

$$\begin{aligned} G_2(f) &= \frac{1}{9} (10 f(x_0) + 16 f(x_1) + 10 f(x_2)) \\ &= \frac{1}{9} (10 \cdot \sqrt{4 - \frac{12}{5}} + 16 \cdot 2 + 10 \cdot \sqrt{4 - \frac{12}{5}}) \\ &= \frac{1}{9} (20 \sqrt{\frac{8}{5}} + 32) \\ &\approx 6.3665 \end{aligned}$$

c) Richardson 4 :

$$\begin{aligned} h_1 = 4 \quad T_1 &= \frac{h_1}{2} (f(-2) + f(2)) = 0 \end{aligned}$$

$$h_2 = 2 \quad T_2 = \frac{h_2}{2} (f(-2) + 2f(0) + f(2)) = 4$$



$$h_k \quad T_k$$

$$4 \quad 0$$

$$2 \quad 4 \quad \rightarrow \quad 4 + \frac{4 - 0}{\left(\frac{4}{2}\right)^2 - 1} = 4 + \frac{4}{4 - 1} = \frac{16}{3}$$

$$\Rightarrow R_4(f) = \frac{16}{3}$$